

Abhishek Singh

📍 E152FD, London ✉ abhisteam@gmail.com ☎ +44-7849780699 in Portfolio ✉ References

Education

Queen Mary University of London <i>MSc in Machine Learning & Computer Vision</i>	<i>London, UK</i> <i>2022 - 2023</i>
◦ Relevant Coursework: Machine Learning, Deep Learning & Computer Vision, Data Mining	
Malaviya National Institute of Technology, Jaipur <i>BTech in Electrical Engineering</i>	<i>Jaipur, India</i> <i>2017 - 2021</i>
◦ Relevant Coursework: Signal Processing, Control Systems, Neural Networks & Graphs	

Skills & Interests

- **Programming & Scripting:** Python, R, Perl, Java, SQL, Bash, Linux
- **Frameworks & Libraries:** PyTorch, TensorFlow, Apache Spark, Flask, React, Node.js, MATLAB, Biopython, Chimera, PyMOL
- **DevOps & Tools:** Docker, Kubernetes, Git, Azure DevOps, Hadoop, Jira, Scrum
- **ML & AI:** Machine Learning, Deep Learning, Computer Vision, Diffusion Models, LLMs, NLP, Classification, Anomaly Detection, Deployment
- **Data Skills:** Data Mining, Big Data Analytics, EDA, Visualization (Matplotlib, Seaborn)
- **Cloud & Platforms:** AWS (SageMaker, S3, Bedrock), Hugging Face (diffusers)
- **Languages:** English (Native), Japanese (N4, working toward N3), Hindi

Experience

AI Drug Discovery <i>Hummingbird Bioscience</i>	<i>Singapore</i> <i>July 2024 - Present</i>
◦ Engineered and deployed ML pipelines for nanobody design using RFDiffusion for backbone generation and ESM-3 for inverse folding, translating research into reproducible production workflows, leveraging Docker containerization, and Linux-based HPC clusters.	
◦ Designed and set up workflows for fine-tuning DiffAb (a diffusion-based model) for generating CDR3 loops targeting CD3 and explored strategies for RFDiffusion adaptation, with detailed documentation to ensure reproducibility in innovative bioinformatics solutions.	
◦ Conducted Molecular Dynamics (MD) simulations using GROMACS on clusters using slurm schedulers to validate designs' physical properties and dynamics, ensuring robustness prior to experimental testing.	
Data Science Analyst <i>Yori Shoreditch</i>	<i>London, UK</i> <i>Dec 2023 - Apr 2024</i>
◦ Enhanced sales forecasting accuracy by building predictive models.	
◦ Ensured high-quality data inputs by cleaning and preparing datasets using SQL and Pandas, uncovered actionable insights by conducting exploratory data analysis with Pandas and Matplotlib, and executed A/B testing to validate model configurations and optimize feature selection.	
Computer Vision Engineer (Contract) <i>Sibylline Advisory</i>	<i>London, UK</i> <i>Nov 2023 - Mar 2024</i>
◦ Engineered a high-accuracy crop quality analysis system, achieving over 95% accuracy by developing a disease classification and detection model using AWS Rekognition and YOLOv5, and implemented a robust CI/CD pipeline with Docker containerization to streamline deployment and scaling, thereby optimizing stakeholder decision-making and boosting agricultural sales through actionable insights.	
Software Consultant <i>Jantacare Services</i>	<i>Noida, India</i> <i>Jan 2022 - Jul 2022</i>
◦ Acted as a techno-functional consultant on Tivoli and IBM Maximo, developing automation scripts using python, Java, performing API calls, and conducting rigorous User Acceptance Testing (UAT).	

Leadership & Projects

Dissertation Project <i>Queen Mary University</i>	<i>London, UK</i> <i>2022 - 2023</i>
◦ Surpassed state-of-the-art in fashion garment synthesis by developing a novel stable diffusion pipeline that generates high-quality garments from text and sketch prompts, outperforming existing models on metrics like FID, CLIP, and structural similarity. Available here .	